Mathematical and Computational Methods

Unit 1: Foundations: numbers, sets and functions

This opening unit assembles the vocabulary and objects on which the whole module rests: the number systems we compute in, the set, interval and summation notation we write with, and—above all—the idea of a *function* together with the catalogue of elementary functions used throughout the course. The aim is fluency with the language, not difficulty: almost everything here will be used in every later unit.

Where this fits. Nothing is assumed from a previous unit; this is the starting point. The chain of number systems ends at the complex numbers $\mathbb{C}$, which we build in Units 2–3, and the exponential, trigonometric and hyperbolic functions introduced here reappear there in the polar and Euler forms. The notions of domain, monotonicity and symmetry set up the calculus of Units 7–10, and the ReLU and sigmoid functions return in the optimisation and data-science material.

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Learning objectives

By the end of this unit you will be able to:

- Use set, interval and Σ/Π notation, and solve inequalities.
- Find a function's maximal domain and range, and compose and decompose functions.
- Classify functions (injective/surjective/bijective, even/odd/periodic) and find inverses.
- Sketch and transform the elementary functions, including $|x|$, sgn, H and ReLU.
- State $\cosh^2 x - \sinh^2 x = 1$ and read the sigmoid $\sigma(x)$ as a probability.

1 Numbers, sets and notation

1.1 Number systems

We begin with the *natural numbers* used for counting and enlarge the number system each time an operation leads outside it.

We use a little set notation straight away; §1.4 sets it out properly. Here $x \in A$ means that x is an element of the set A, $\{\, x \;:\; P(x) \,\}$ is the set of all x with the property $P(x)$, and $A \subseteq B$ means that every element of A also lies in B; we write $A \subset B$ when, in addition, $A \neq B$.

Definition 1.1 (The standard number systems).

- $\mathbb{N} = \{\, 1, 2, 3, \dots \,\}$ — the *natural numbers* (counting). We write $\mathbb{N}_0 = \{\, 0, 1, 2, \dots \,\}$ when 0 is included.
- $\mathbb{Z} = \{\, \dots, -2, -1, 0, 1, 2, \dots \,\}$ — the *integers*; closed under subtraction (the difference of two integers is always an integer).
- $\mathbb{Q} = \left\{ \dfrac{p}{q} \;:\; p \in \mathbb{Z},\ q \in \mathbb{N} \right\}$ — the *rationals*; closed under division by nonzero rationals.
- $\mathbb{R}$ — the *real numbers*; informally the "number line with no gaps" (this completeness is made precise in a later university course on Analysis, not in this module). It contains $\mathbb{Q}$ together with numbers such as $\sqrt{2}$, π and e that are not fractions.
- $\mathbb{C} = \{\, a + b\,\mathrm{i} \;:\; a, b \in \mathbb{R} \,\}$, where i is a symbol with $\mathrm{i}^2 = -1$ — the *complex numbers*, the subject of Units 2–3.

These sit inside one another, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

Remark. Each extension removes an obstruction. Subtraction can leave $\mathbb{N}$ (e.g. $3 - 5$), so we pass to $\mathbb{Z}$; division can leave $\mathbb{Z}$ (e.g. $1 \div 2$), so we pass to $\mathbb{Q}$; taking limits or roots can leave $\mathbb{Q}$ (e.g. $\sqrt{2}$), so we pass to $\mathbb{R}$; and the equation $x^2 + 1 = 0$ has no real solution, which forces the step to $\mathbb{C}$.

1.2 Rational and irrational numbers

A real number is *rational* if it can be written as a fraction of integers and *irrational* otherwise. A second, independent classification cuts across this one.

Definition 1.2 (Algebraic and transcendental numbers). A real (or complex) number is *algebraic* if it is a root of some nonzero polynomial with integer coefficients, and *transcendental* otherwise.

Every rational is algebraic (p/q is a root of $qx - p$), and so are some irrationals: $\sqrt{2}$ is a root of $x^2 - 2$. Numbers such as π and e are transcendental, the root of *no* integer polynomial, and so they are irrational too. The next example shows the standard way to prove a number irrational.

Example 1.1 ($\sqrt{2}$ *is irrational).* Suppose, for contradiction, that $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{N}$ sharing no common factor. Squaring gives $p^2 = 2q^2$, so p^2 is even, hence p is even (the square of an odd number is odd), say $p = 2k$. Then $4k^2 = 2q^2$, i.e. $q^2 = 2k^2$, so q^2 is even and q is even too. But then p and q share the factor 2, contradicting our assumption. Therefore no such fraction exists and $\sqrt{2}$ is irrational.

1.3 Intervals and infinity

For $a < b$ in $\mathbb{R}$ we use the standard interval notation

$$(a, b) = \{\, x \in \mathbb{R} \;:\; a < x < b \,\}, \qquad [a, b] = \{\, x \in \mathbb{R} \;:\; a \le x \le b \,\},$$

with the half-open intervals $[a, b)$ and $(a, b]$ defined analogously: a square bracket includes the endpoint, a round bracket excludes it. Unbounded intervals use the symbols $\pm\infty$, for instance

$$[a, \infty) = \{\, x \in \mathbb{R} \;:\; x \ge a \,\}, \qquad (-\infty, b) = \{\, x \in \mathbb{R} \;:\; x < b \,\}.$$

Note. The symbol ∞ is *not* a real number. In interval notation it marks an end with no bound: $[a, \infty)$ contains every real number from a upwards. An interval boundary at ∞ is therefore always written with a round bracket, never a square one.

1.4 Sets and set-builder notation

A *set* is a collection of distinct objects. Small sets are written by listing their elements, as in $\{1, 2, 3\}$; larger or infinite sets are described by a defining property using *set-builder notation*,

$$\{\, x \;:\; P(x) \,\} \qquad \text{“the set of all } x \text{ such that } P(x) \text{ holds”}.$$

For example $\{\, x \in \mathbb{R} \;:\; x^2 < 4 \,\} = (-2, 2)$. The basic operations are union $A \cup B$ (elements in A *or* B), intersection $A \cap B$ (in A *and* B), difference $A \setminus B$ (in A but not B), and the empty set $\emptyset$ containing no elements.

1.5 Sum and product notation

Long sums and products are abbreviated with the Greek capitals Σ (sigma) and Π (pi):

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n, \qquad \prod_{k=1}^{n} a_k = a_1 \cdot a_2 \cdots a_n.$$

Here k is a *dummy index* running over the stated range. A sum splits term by term and constants come out, $\sum(\alpha a_k + \beta b_k) = \alpha \sum a_k + \beta \sum b_k$, and $\sum_{k=1}^{n} 1 = n$. Two facts worth remembering are the arithmetic-series sum and the factorial,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \prod_{k=1}^{n} k = n!\,.$$

Example 1.2 ***(Evaluating a sum).*** The sum of the first five odd numbers is

$$\sum_{k=1}^{5}(2k-1) = 1+3+5+7+9 = 25.$$

Splitting the sum confirms this in general:

$$\sum_{k=1}^{n}(2k-1) = 2\sum_{k=1}^{n}k - \sum_{k=1}^{n}1 = 2\cdot\frac{n(n+1)}{2} - n = n^2,$$

and indeed $5^2 = 25$.

1.6 Inequalities

Inequalities obey rules close to those for equations, with one crucial exception.

Note (Rules for inequalities). For real numbers one may add or subtract the same quantity on both sides freely, and multiply or divide both sides by a *positive* number. Multiplying or dividing by a *negative* number **reverses** the inequality sign. Also, for $a > 0$ and the absolute value $|x|$ (the distance from x to 0; see §3.5),

$$|x| \le a \iff -a \le x \le a, \qquad |x| \ge a \iff x \le -a \text{ or } x \ge a.$$

For $a, b \ge 0$, $a < b \iff a^2 < b^2$, because $b^2 - a^2 = (b-a)(b+a)$ and $b+a > 0$ unless $a = b = 0$. So an inequality between two non-negative quantities, such as two moduli, may be squared: $\sqrt{x} < 3 \iff x < 9$ for $x \ge 0$. Without the sign condition squaring fails: $-3 < 2$, but $9 > 4$.

Example 1.3 ***(A quadratic and an absolute-value inequality).*** To solve $x^2 - x - 6 < 0$, factor: $(x-3)(x+2) < 0$. The product is negative exactly between the roots, so the solution is $-2 < x < 3$, i.e. the interval $(-2, 3)$. A quotient $(x-a)/(x-b)$ has the same sign as the product $(x-a)(x-b)$ wherever it is defined, since it equals $(x-a)(x-b)/(x-b)^2$ and $(x-b)^2 > 0$; so $(x-3)/(x+2) < 0$ has the same solution $(-2, 3)$.

To solve $|2x-1| \le 3$, remove the modulus: $-3 \le 2x - 1 \le 3$, hence $-2 \le 2x \le 4$ and finally $-1 \le x \le 2$, i.e. $[-1, 2]$.

2 Functions and their properties

2.1 What a function is

Definition 2.1 (Function, domain, codomain, range). A *function* $f : A \to B$ is a rule assigning to *each* element $x \in A$ exactly one element $f(x) \in B$. The set A is the *domain*, B is the *codomain*, and the *range* (or image) is the set of values actually attained,

$$f(A) = \{\, f(x) \;:\; x \in A \,\} \subseteq B.$$

When a function is given only by a formula, its *maximal* (or natural) domain is the largest set of real inputs for which the formula makes sense—excluding, for instance, division by zero or even roots of negative numbers.

Example 2.1 ***(Maximal domain).*** For $f(x) = \dfrac{\sqrt{x+3}}{x-1}$ we need $x + 3 \geq 0$ (for the square root) and $x - 1 \neq 0$ (no division by zero). The maximal domain is therefore

$$\{\, x \in \mathbb{R} \;:\; x \geq -3,\; x \neq 1 \,\} = [-3, 1) \cup (1, \infty).$$

Example 2.2 ***(Finding a range).*** For $g(x) = 2 - \sqrt{4 - x}$ the maximal domain is $x \leq 4$. As x runs over $(-\infty, 4]$ the inner term $\sqrt{4 - x}$ takes every value in $[0, \infty)$, so $-\sqrt{4 - x}$ ranges over $(-\infty, 0]$ and $g(x) = 2 - \sqrt{4 - x}$ ranges over $(-\infty, 2]$. The maximum value 2 is attained at $x = 4$, so the range is $(-\infty, 2]$.

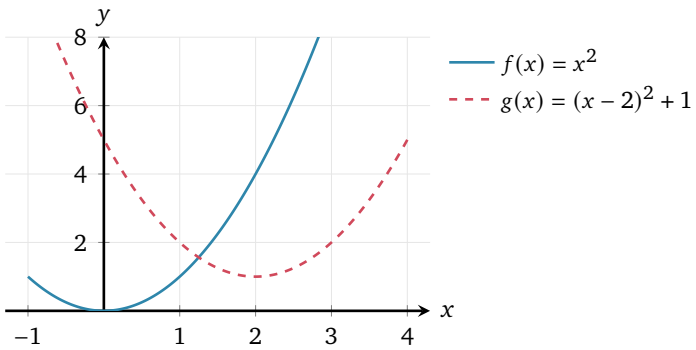
Note. A formula defines a function only if it gives a *single* output for each input. Graphically this is the *vertical line test*: every vertical line meets the graph at most once.

2.2 Graphs and graph transformations

The *graph* of f is the set of points $\{\, (x, f(x)) \;:\; x \in A \,\}$ in the plane. Once the graph of a function f is known, the graphs of simple algebraic modifications follow by shifting, stretching and reflecting.

Transformation	Formula	Effect on the graph
Vertical shift	$f(x) + c$	up by c (down if $c < 0$)
Horizontal shift	$f(x - c)$	right by c (left if $c < 0$)
Vertical scaling	$a\,f(x)$	by factor $\lvert a \rvert$ from the x-axis, and reflected in the x-axis too if $a < 0$
Horizontal scaling	$f(bx)$	by factor $1/b$ (for $b > 0$): compressed towards the y-axis if $b > 1$, stretched if $0 < b < 1$
Reflection in x-axis	$-f(x)$	flip top-to-bottom
Reflection in y-axis	$f(-x)$	flip left-to-right

For instance the graph of $g(x) = (x - 2)^2 + 1$ is the parabola $y = x^2$ shifted two units right and one unit up:



2.3 Composition of functions

Definition 2.2 (Composition). Given f and g, the *composite* $f \circ g$ is defined by $(f \circ g)(x) = f\big(g(x)\big)$: first apply g, then f. Its domain consists of those x in the domain of g for which $g(x)$ lies in the domain of f.

Composition is generally *not* commutative: $f \circ g$ and $g \circ f$ usually differ.

Example 2.3 *(Composition, and decomposition).* Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$. Then

$$(f \circ g)(x) = \sqrt{1 - x^2} \quad \text{(domain } [-1, 1]\text{)}, \qquad (g \circ f)(x) = 1 - x \quad \text{(domain } x \geq 0\text{)},$$

so the order matters, and so does the domain. Conversely, recognising a function as a composite is a key skill: $h(x) = (2x + 1)^5$ is $F \circ G$ with $G(x) = 2x + 1$ and $F(u) = u^5$. This inner–outer decomposition is exactly what the chain rule of Unit 7 needs.

2.4 Injective, surjective and bijective functions

Definition 2.3 (Three properties of a function $f : A \to B$). f is *injective* (one-to-one) if different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. It is *surjective* (onto) if its range is all of B. It is *bijective* if it is both injective and surjective.

Note. Injectivity is the *horizontal line test*: every horizontal line meets the graph at most once.

Example 2.4 *(The same rule, different verdicts).* As a map $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is *not* injective ($f(-1) = f(1) = 1$) and *not* surjective (it never takes negative values). Changing the domain or the codomain changes the verdicts one at a time:

$f(x) = x^2$ as a map	injective?	surjective?
$\mathbb{R} \to \mathbb{R}$	no	no
$\mathbb{R} \to [0, \infty)$	no	yes
$[0, \infty) \to \mathbb{R}$	yes	no
$[0, \infty) \to [0, \infty)$	yes	yes: bijective

Shrinking the domain to $[0, \infty)$ removes the competing inputs x and $-x$; shrinking the codomain to $[0, \infty)$ leaves only values that $x = \sqrt{y}$ reaches. Injectivity, surjectivity and bijectivity are properties of the rule *together with* its stated domain and codomain.

2.5 Inverse functions and monotonicity

Definition 2.4 (Inverse function). If $f : A \to B$ is bijective, its *inverse* $f^{-1} : B \to A$ is the function that undoes it: $f^{-1}(y) = x$ exactly when $f(x) = y$. Equivalently $f^{-1}\big(f(x)\big) = x$ and $f\big(f^{-1}(y)\big) = y$. Only a bijection has an inverse: a second input with the same output, or an output never reached, would leave f^{-1} undefined there.

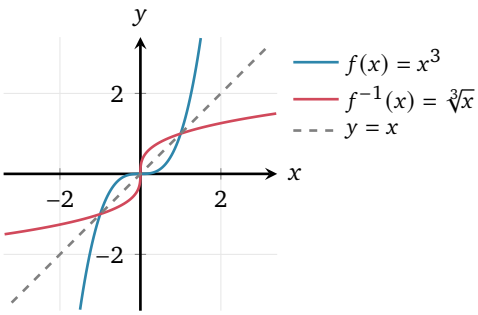
A function is *strictly increasing* if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$, and *strictly decreasing* if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$; either way it is *strictly monotonic*. A strictly monotonic function is automatically injective; restricting its codomain to its range then makes it bijective, so it has an inverse on that range—the most common way to guarantee one exists. To find an inverse algebraically, write $y = f(x)$ and solve for x in terms of y.

Example 2.5 ***(Finding an inverse).*** Let $f(x) = \sqrt{x-1}$ on the domain $[1, \infty)$, with range $[0, \infty)$. Setting $y = \sqrt{x-1}$ and solving gives $x = y^2 + 1$, so

$$f^{-1}(x) = x^2 + 1, \qquad \text{domain } [0, \infty), \ \text{range } [1, \infty).$$

Note how domain and range swap between f and f^{-1}.

Note. The graph of f^{-1} is the mirror image of the graph of f in the line $y = x$, because reflecting in $y = x$ swaps the roles of input and output. The figure below shows this for $f(x) = x^3$, which is bijective on all of $\mathbb{R}$: solving $y = x^3$ gives $x = \sqrt[3]{y}$, so $f^{-1}(x) = \sqrt[3]{x}$ for every real x (no domain restriction needed). Write $\sqrt[3]{x}$, not $x^{1/3}$: powers a^x are defined only for $a > 0$, and a computer asked for $(-8)^{1/3}$ may return a complex number rather than -2.



2.6 Symmetry: even, odd and periodic functions

Definition 2.5 (Symmetry). A function with a domain symmetric about 0 is *even* if $f(-x) = f(x)$ for all x (graph symmetric about the y-axis) and *odd* if $f(-x) = -f(x)$ for all x (graph symmetric about the origin). A function is *periodic* with period $T > 0$ if $f(x + T) = f(x)$ for all x; the smallest such T, when there is one, is *the* period.

For example x^2 and $\cos x$ are even, x^3 and $\sin x$ are odd, and $\sin x$ is periodic with period 2π. Most functions are neither even nor odd. Recognising symmetry halves the work of sketching a graph and, later, of computing integrals.

3 Elementary functions and their graphs

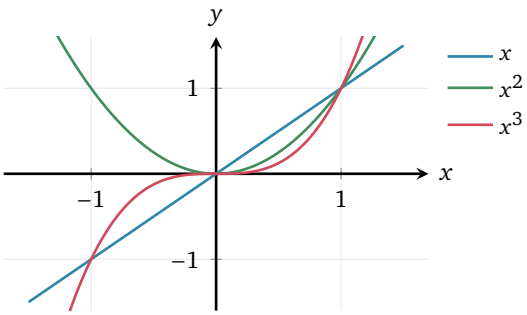
The functions below, with their compositions, inverses and the four arithmetic operations, make up the *elementary functions*—the working vocabulary for the rest of the module.

3.1 Power, polynomial and rational functions

A *power function* is $x \mapsto x^n$; in this unit n is a positive integer (Unit 7 allows any real n). It is even when n is even (symmetric, non-negative) and odd when n is odd. A *polynomial* of degree n is a finite sum

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0,$$

with domain all of $\mathbb{R}$; for large $|x|$ its sign and growth are governed by the leading term $a_n x^n$. A *rational function* is a quotient $p(x)/q(x)$ of two polynomials; its maximal domain excludes the zeros of the denominator q, near which the graph typically has vertical asymptotes.



Example 3.1 ***(Domain of a rational function).*** For

$$r(x) = \frac{x+1}{x^2-4} = \frac{x+1}{(x-2)(x+2)}$$

the denominator vanishes at $x = \pm 2$, so the maximal domain is $\mathbb{R} \setminus \{-2, 2\}$. The numerator is not zero at either point, so $|r(x)|$ grows without bound there and the lines $x = -2$ and $x = 2$ are vertical asymptotes. (Where numerator and denominator vanish together, as for $(x-2)/(x^2-4)$ at $x = 2$, the graph may have a gap instead.)

3.2 Exponential and logarithmic functions

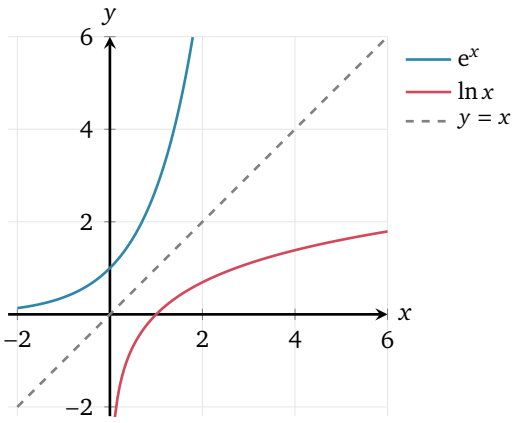
For a base $a > 0$, $a \neq 1$, the *exponential function* a^x has domain $\mathbb{R}$ and range $(0, \infty)$, and obeys the index laws

$$a^{x+y} = a^x a^y, \qquad (a^x)^y = a^{xy}, \qquad a^0 = 1, \qquad a^{-x} = \frac{1}{a^x}.$$

The most important base is Euler's number $\mathrm{e} \approx 2.71828$, giving the *natural exponential* e^x. Its inverse is the *natural logarithm* $\ln = \log_{\mathrm{e}}$, with domain $(0, \infty)$, range $\mathbb{R}$, and the laws

$$\ln(xy) = \ln x + \ln y, \quad \ln\frac{x}{y} = \ln x - \ln y, \quad \ln(x^r) = r \ln x, \quad \log_a x = \frac{\ln x}{\ln a}.$$

Because $\ln$ inverts e^x, the two graphs are reflections of one another in the line $y = x$.

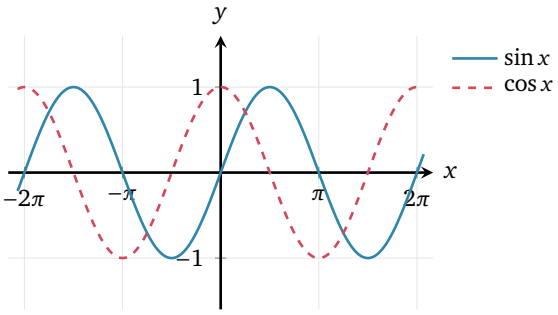


Example 3.2 ***(Solving with exponentials and logarithms).*** To solve $2^x = 10$, take natural logs: $x \ln 2 = \ln 10$, so $x = \dfrac{\ln 10}{\ln 2} = \log_2 10 \approx 3.32$. The log laws also simplify expressions directly, e.g. $\ln 8 - \ln 2 = \ln \frac{8}{2} = \ln 4$.

3.3 Trigonometric and inverse-trigonometric functions

At university angles are measured in *radians*; recall $x^\circ = \frac{\pi}{180} x$ radians, so $180^\circ = \pi$. The functions sin and cos have domain $\mathbb{R}$, range $[-1, 1]$ and period 2π; sin is odd and cos is even. The standard values are worth memorising.

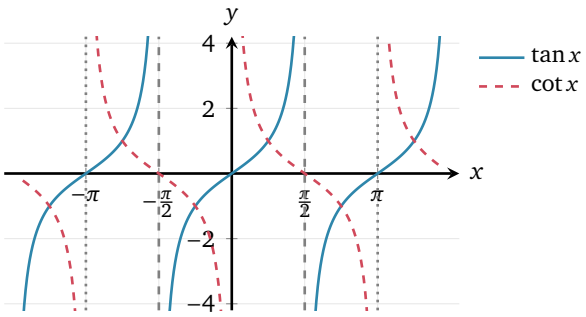
θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0



The remaining trigonometric functions are built from sin and cos. The *tangent* and *cotangent* are

$$\tan x = \frac{\sin x}{\cos x}, \qquad \cot x = \frac{\cos x}{\sin x},$$

alongside the reciprocals $\sec x = 1/\cos x$ and $\csc x = 1/\sin x$; $\cot x$ equals $1/\tan x$ wherever both are defined. Both tan and cot are odd, take every real value (range $\mathbb{R}$) and are periodic with period π. Their domains exclude the zeros of the denominator: tan is undefined where $\cos x = 0$, i.e. at $x = \frac{\pi}{2} + k\pi$, and cot is undefined where $\sin x = 0$, i.e. at $x = k\pi$ (with $k \in \mathbb{Z}$); at each excluded point the graph has a vertical asymptote (the grey dashed lines belong to tan, the grey dotted ones to cot).



We will lean on a handful of identities, recapped here for reference and used heavily in Units 2–3:

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta,$$

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta,$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta, \qquad \cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$

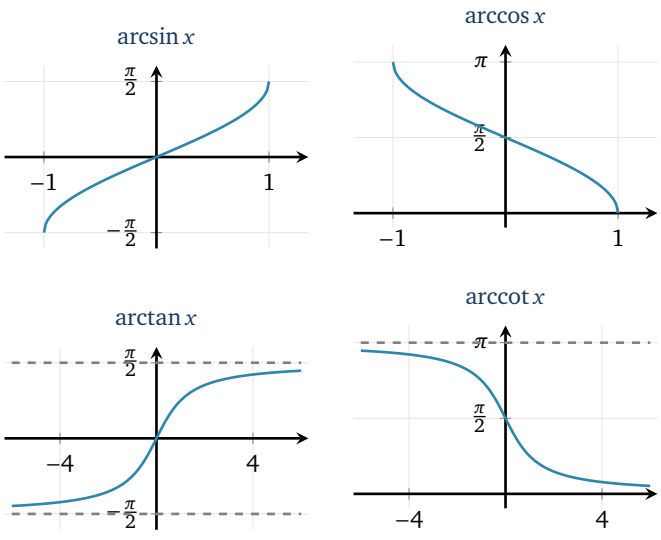
Inverse trigonometric functions

Being periodic, the trigonometric functions are far from injective, so each of the four we need—sin, cos, tan and cot—is made invertible by restricting it to a single monotonic *principal* branch. This gives the inverse functions

$$\arcsin\colon [-1,1] \to [-\tfrac{\pi}{2}, \tfrac{\pi}{2}], \qquad \arccos\colon [-1,1] \to [0,\pi],$$

$$\arctan\colon \mathbb{R} \to (-\tfrac{\pi}{2}, \tfrac{\pi}{2}), \qquad \operatorname{arccot}\colon \mathbb{R} \to (0,\pi),$$

where arcsin inverts sin on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, arccos inverts cos on $[0,\pi]$, arctan inverts tan on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and arccot inverts cot on $(0,\pi)$. The two tangent-type inverses are bounded: arctan has horizontal asymptotes $y = \pm\frac{\pi}{2}$, and arccot has $y = 0$ and $y = \pi$. As with any inverse, the graph of each is the reflection of its restricted original in the line $y = x$; all four are drawn below.



Example 3.3 ***(Trigonometric equations).*** Solve $\cos x = \frac{1}{2}$ for $x \in [0, 2\pi)$. From the table $\arccos\frac{1}{2} = \frac{\pi}{3}$, and since cosine is even and 2π-periodic the solutions in this interval are $x = \frac{\pi}{3}$ and $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$.

For sine the partner comes from $\sin(\pi - \theta) = \sin\theta$ instead: $\sin x = \frac{\sqrt{3}}{2}$ has principal value

$\arcsin\frac{\sqrt{3}}{2} = \frac{\pi}{3}$ and partner $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$, and these are its two solutions in $[0, 2\pi)$.

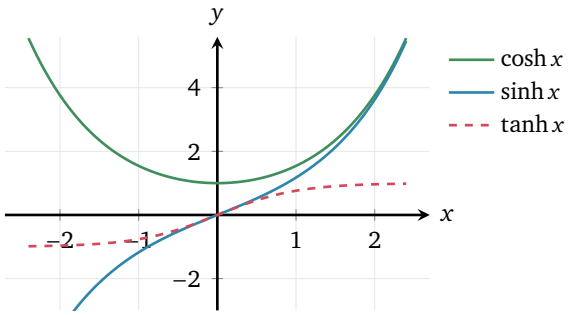
Example 3.4 ***(Tangent and its inverse).*** Solve $\tan x = 1$ for $x \in [0, 2\pi)$. The principal value is $\arctan 1 = \frac{\pi}{4}$, and since tan has period π the solutions in this interval are $x = \frac{\pi}{4}$ and $x = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$. Note the distinction: $\arctan 1$ returns only the single principal value in $(-\frac{\pi}{2}, \frac{\pi}{2})$, whereas the equation $\tan x = 1$ has infinitely many solutions.

3.4 Hyperbolic functions (brief)

Built from the exponential, the *hyperbolic* functions are

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}.$$

Here sinh is odd and cosh is even, $\cosh x \geq 1$ for all x, and tanh is odd with horizontal asymptotes $y = \pm 1$. They satisfy the identity below, the hyperbolic analogue of $\sin^2 + \cos^2 = 1$.



Example 3.5 ***(The fundamental identity).*** Directly from the definitions,

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4} = \frac{4\,e^x e^{-x}}{4} = 1.$$

Remark. We keep hyperbolic functions brief here. Their deeper meaning—and the reason cosh and cos look so alike—emerges in Unit 3, where Euler's formula gives $\cos(ix) = \cosh x$. They reappear once more as standard derivatives in Unit 7.

3.5 Piecewise-defined functions

A function may be specified by different formulas on different parts of its domain. Several such functions are central to physics and to data science.

Definition 3.1 (Four piecewise functions).

$$\text{absolute value:}\quad |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases} \qquad \text{sign:}\quad \operatorname{sgn}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0, \end{cases}$$

$$\text{Heaviside step:}\quad H(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad \text{ReLU:}\quad \operatorname{ReLU}(x) = \max(0, x).$$

Example 3.6 ***(Reading a piecewise definition).*** $\operatorname{ReLU}(x) = \max(0, x)$ reads piecewise as 0 for $x < 0$ and x for $x \geq 0$. To check an identity between piecewise functions, split at the breakpoints and check each piece. For instance $x \operatorname{sgn}(x) = |x|$: for $x > 0$ both sides are x; at $x = 0$ both are 0; and for $x < 0$ the left side is $x \cdot (-1) = -x = |x|$. The same split shows that ReLU has the closed form $\operatorname{ReLU}(x) = \frac{1}{2}(x + |x|)$.

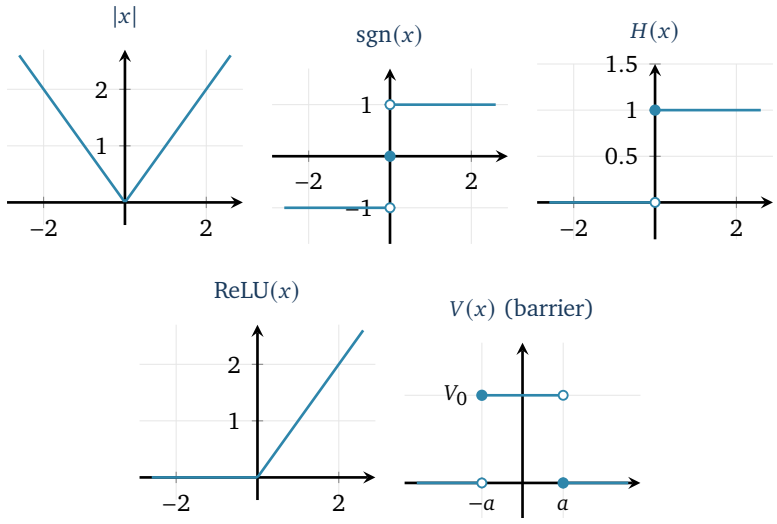
The physics students should meet one more piecewise function: the *rectangular potential barrier*, which is constant on a finite interval and zero outside it.

Definition 3.2 (Rectangular potential barrier). For a height $V_0 > 0$ and width $2a > 0$,

$$V(x) = \begin{cases} V_0, & -a \leq x < a, \\ 0, & \text{otherwise.} \end{cases}$$

The values exactly at $x = \pm a$ are a convention with no physical meaning. This is the finite square barrier of quantum mechanics: a particle of energy below V_0 has a nonzero probability of *tunnelling* through it, which classical mechanics forbids. It is also a basic building block for assembling more complicated potentials out of simple pieces.

The five functions are drawn below (taking $V_0 = 1$, $a = 1$ for the barrier). A filled dot marks an included endpoint and an open dot an excluded one, so the jumps in sgn, H and V are unambiguous.



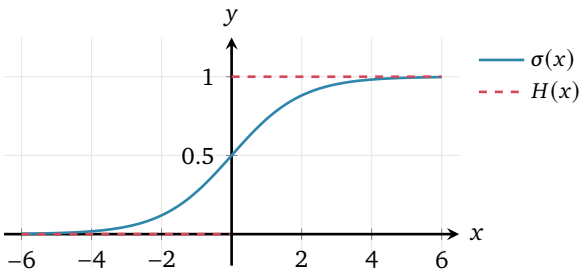
3.6 The logistic (sigmoid) function

The *logistic* (or *sigmoid*) function is *not* piecewise: it is a single smooth formula. It is a smooth cousin of the step H and one of the most important functions in data science.

Definition 3.3 (The logistic (sigmoid) function).

$$\sigma(x) = \frac{1}{1 + \mathrm{e}^{-x}}.$$

It is smooth and strictly increasing, with $\sigma(0) = \frac{1}{2}$, range $(0, 1)$ and limits $\sigma(x) \to 1$ as $x \to \infty$ and $\sigma(x) \to 0$ as $x \to -\infty$; it is thus a smooth approximation of the Heaviside step H, and it has the symmetry $\sigma(-x) = 1 - \sigma(x)$.



A probability from any number. Because σ sends every real number into the open interval $(0, 1)$, its output can be read as a *probability* —which is what makes it ubiquitous in data science. In *logistic regression*, a real-valued score s assembled from the data (typically a weighted sum of measured features) is turned into an estimated probability $\sigma(s)$ that an item belongs to a class: a large positive s gives a probability close to 1, a large negative s one close to 0, and $s = 0$ the noncommittal $\frac{1}{2}$. The symmetry $\sigma(-x) = 1 - \sigma(x)$ is exactly the statement that the probabilities of the two classes sum to 1.

Remark. ReLU and σ are the two best-known *activation functions* in neural networks: the data-science students will meet them again as the nonlinearities between layers. We will differentiate σ in Unit 7, finding the tidy fact $\sigma'(x) = \sigma(x)\big(1 - \sigma(x)\big)$, which Unit 8 then puts to use.

Summary

Concept	Key idea
Number systems	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; each extension repairs an operation that escapes the previous system.
Algebraic vs transcendental	Algebraic = root of an integer polynomial (every rational, and some irrationals like $\sqrt{2}$); transcendental (π, e) otherwise.
Notation	Intervals $[a,b]$, (a,b); set-builder $\{x : P(x)\}$; sums/products $\sum$, $\prod$; inequality rules (a negative multiplier flips the sign).
Function	A rule $f : A \to B$, one output per input; domain, codomain, range, maximal domain.
Transformations	$f(x)+c$, $f(x-c)$, $af(x)$, $f(bx)$, $-f(x)$, $f(-x)$ shift, stretch and reflect the graph.
Composition	$(f \circ g)(x) = f(g(x))$; order matters; decomposition prepares the chain rule.
Inj./surj./bij.	One-to-one / onto / both; bijective $\iff$ invertible; strict monotonicity $\Rightarrow$ injective.
Inverse	Solve $y = f(x)$ for x, then rename; domain and range swap; graph reflects in $y = x$.
Symmetry	Even $f(-x) = f(x)$, odd $f(-x) = -f(x)$, periodic $f(x+T) = f(x)$.
Elementary functions	Power/polynomial/rational, e^x and ln, the trigonometric and inverse-trig functions, a brief look at sinh, cosh, tanh, the piecewise $\lvert x\rvert$, sgn, H, ReLU, and the smooth sigmoid σ.

Next unit: the final step in the number chain—the complex numbers $\mathbb{C}$—their algebra, the Argand diagram, and the polar form.